



Physical Constants

Coulomb constant $k = \frac{1}{4\pi\epsilon_0} = 8.987 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$

Electron mass $= 9.109 \times 10^{-31} \text{ kg}$

Electron charge $= 1.602 \times 10^{-19} \text{ C}$

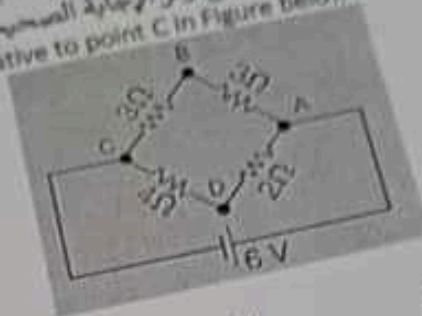
Acceleration due to gravity $g = 9.80 \text{ m/s}^2$

Permittivity of free space $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N}\cdot\text{m}^2$

Proton mass $= 1.672 \times 10^{-27} \text{ kg}$

The mathematical constant $\pi = 3.14$

Q1) What is the potential of point B relative to point C in Figure below?



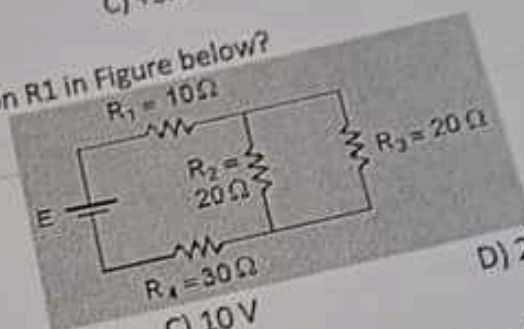
A) +6.0 V

B) +4.0 V

C) +3.0 V

D) +2.0 V

Q.2) If $E = 40 \text{ V}$, what is the voltage on R_1 in Figure below?



A) 6.7 V

B) 8.0 V

C) 10 V

D) 20 V

Q.3) A $2.0 \mu\text{F}$ capacitor is charged through a $50 \text{ k}\Omega$ resistor. How long does it take for the capacitor to reach 90% of full charge?

A) 0.923 s

B) 0.23 s

C) 0.16 s

D) 2.92 s

Q.4) An electron follows a circular path (radius = 15 cm) in a uniform magnetic field of magnitude $(3 \times 10^{-4} \text{ T})$. What is the period of this motion?

A) $0.12 \mu\text{s}$

B) 1.6 ms

C) $0.33 \mu\text{s}$

D) 3.8 ms

Q.5) A particle ($q = -4.0 \mu\text{C}$, $m = 5.0 \text{ mg}$) moves in a uniform magnetic field with a velocity of 2.0 km/s and a direction that is 50° away from that of the magnetic field. The particle experiences an acceleration with a magnitude of 5.8 m/s^2 . What is the magnitude of the magnetic field?

A) 5.3 T

B) 3.6 T

C) 7.1 T

D) 4.7 T

Q.6) A light bulb is rated at 30 W when operated at 120 V . How much charge enters the bulb in 1.0 min ?

B) 15 C

C) 14 C

D) 13 C

- Q.9) By what factor is the capacitance of a metal sphere multiplied if its volume is tripled?
 A) 3 B) $(3)^{-\frac{1}{3}}$ C) $(3)^{\frac{1}{3}}$ D) $\frac{1}{3}$
- Q.10) When a battery is connected to the plates of a $3.00 \mu\text{F}$ capacitor, it stores a charge of $27.0 \mu\text{C}$. What is the voltage of the battery?
 A) 3 V B) 12 V C) 6 V D) 9 V
- Q.11) An air-filled spherical capacitor is constructed with inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. Calculate the capacitance of the device.
 A) 15.6 pF B) 12.2 pF C) 6.3 pF D) 9.7 pF
- Q.12) The electric potential at $x = 3.00 \text{ m}$ is 120 V, and the electric potential at $x = 5.00 \text{ m}$ is 190 V. What is the x component of the electric field in this region, assuming the field is uniform?
 A) +17.5 N/C B) -17.5 N/C C) +35.0 N/C D) -35.0 N/C
- Q.13) A proton is released from rest at the origin in a uniform electric field in the positive x direction with magnitude 850 N/C. What is the change in the electric potential energy of the proton-field system when the proton travels to $x = 2.50 \text{ m}$?
 A) $3.4 \times 10^{-16} \text{ J}$ B) $+3.4 \times 10^{-16} \text{ J}$ C) $-2.2 \times 10^{-16} \text{ J}$ D) $+2.2 \times 10^{-16} \text{ J}$
- Q.14) Object A has a charge of $+2 \mu\text{C}$, and object B has a charge of $+6 \mu\text{C}$. Which statement is true about the electric forces on the objects?
 A) $F_{AB} = -F_{BA}$ B) $F_{AB} = 0.5F_{BA}$ C) $F_{AB} = 3F_{BA}$ D) $3F_{AB} = F_{BA}$
- Q.15) A charge of $170 \mu\text{C}$ is at the center of a cube of edge 80 cm. No other charges are nearby. Find the flux (N/C $\cdot\text{m}^2$) through the whole surface of the cube.
 A) 1.92×10^7 B) 2.48×10^7 C) 1.92×10^7 D) 7.2×10^7
- Q.16) A 12.0 V battery is connected to a capacitor, resulting in $54.0 \mu\text{C}$ of charge stored on the capacitor. How much energy is stored in the capacitor?
 A) $1.0 \times 10^{-4} \text{ J}$ B) $3.24 \times 10^{-4} \text{ J}$ C) $12.2 \times 10^{-4} \text{ J}$ D) $2.0 \times 10^{-4} \text{ J}$



Q.7) A rod of 2 m length and a square ($2.0 \text{ mm} \times 2.0 \text{ mm}$) cross section is made of a material with a resistivity of $(6 \times 10^{-8} \text{ Ohm} \cdot \text{m})$. If a potential difference of 0.50 V is placed across the ends of the rod, what power generated in the rod?

- A) 3.0 W B) 5.3 W C) 8.3 W D) 16.3 W

Q.8) The radius of 22-gauge Nichrome wire is 0.32 mm. If a potential difference of 10 V is maintained across a 1.0 m length of the Nichrome wire, what is the current in the wire?

- A) 3.2 A B) 7.3 A C) 6.4 A D) 1.34 A
- ($\rho_{\text{Nichrome}} = 1 \times 10^{-6} \text{ Ohm} \cdot \text{meter}$)

Q.9) By what factor is the capacitance of a metal sphere multiplied if its volume is tripled?

- A) 3 B) $(3)^{-\frac{1}{3}}$ C) $(3)^{\frac{1}{3}}$ D) $\frac{1}{3}$

Q.10) When a battery is connected to the plates of a $3.00 \mu\text{F}$ capacitor, it stores a charge of $27.0 \mu\text{C}$. What is the voltage of the battery?

- A) 3 V B) 12 V C) 6 V D) 9 V

Q.11) An air-filled spherical capacitor is constructed with inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. Calculate the capacitance of the device.

- A) 15.6 pF B) 12.2 pF C) 6.3 pF D) 9.7 pF

Q.12) The electric potential at $x = 3.00 \text{ m}$ is 120 V, and the electric potential at $x = 5.00 \text{ m}$ is 190 V. What is the x component of the electric field in this region, assuming the field is uniform?

- A) +17.5 N/C B) -17.5 N/C C) +35.0 N/C D) -35.0 N/C

Q.13) A proton is released from rest at the origin in a uniform electric field in the positive x direction magnitude 850 N/C. What is the change in the electric potential energy of the proton-field system when the proton travels to $x = 2.50 \text{ m}$?

- A) $-3.4 \times 10^{-16} \text{ J}$ B) $+3.4 \times 10^{-16} \text{ J}$ C) $-2.2 \times 10^{-16} \text{ J}$ D) $+2.2 \times 10^{-16} \text{ J}$

Q.14) Object A has a charge of $+2 \mu\text{C}$, and object B has a charge of $+6 \mu\text{C}$. Which statement is true about the electric forces on the objects?

- A) $F_{AB} = -F_{BA}$ B) $F_{AB} = 0.5 F_{BA}$ C) $F_{AB} = 3 F_{BA}$ D) $3 F_{AB}$

Q.15) A charge of $170 \mu\text{C}$ is at the center of a cube of edge 80 cm. No other charges are nearby. What is the magnitude of the electric flux ($\text{N} \cdot \text{m}^2/\text{C}$) through the whole surface of the cube.

- A) 5.2×10^7 B) 2.48×10^7 C) 1.92×10^7 D) 7.2×10^7

Q.16) A 12.0 V battery is connected to a capacitor, resulting in $54.0 \mu\text{C}$ of charge stored on the capacitor. How much energy is stored in the capacitor?

- A) $5.41 \times 10^{-4} \text{ J}$ B) $3.24 \times 10^{-4} \text{ J}$ C) $12.2 \times 10^{-4} \text{ J}$ D) $4.3 \times 10^{-4} \text{ J}$